

Regular Expressions and DFAs

Definition: regular expression

Let Σ be an alphabet. A *regular expression* over Σ is a string over the alphabet $\Sigma \cup \{ (,), \emptyset, \cup, \star \}$ that can be obtained through the application of the following rules:

- $\emptyset$ and e are regular expressions.
- For each $a \in \Sigma$, a is a regular expression.
- If R_1 and R_2 are regular expressions, then so are $(R_1 \cup R_2)$, $(R_1 R_2)$, and $R_1^\star$.

Q1. Let $\Sigma = \{a, b\}$. Let L be the language of all strings in Σ^* in which every occurrence of a is immediately followed by b .

- (a) Give an example of a regular expression α that represents L . Briefly explain why your answer is correct.
- (b) Give a second, structurally different regular expression β that represents L .

Q2. Consider your α and β from Q1.

- (a) Prove that $\mathcal{L}(\alpha) = \mathcal{L}(\beta)$.
- (b) A language can be represented by infinitely many different regular expressions. Explain briefly why this must be true.

Q3. Let α and β be regular expressions over an alphabet Σ . Prove the following:

- (a) $\alpha^* \alpha^* = \alpha^*$
- (b) $(\alpha \cup \beta)^* = (\alpha^* \beta^*)^*$
- (c) $(\alpha \beta)^* \alpha = \alpha (\beta \alpha)^*$

Q4. Let α be a regular expression over Σ . For a string $w = w_1 w_2 \cdots w_n$, define $w^R = w_n \cdots w_2 w_1$. For a language $L \subseteq \Sigma^*$, define $L^R = \{w^R : w \in L\}$.

- (a) Prove that if L is regular, then L^R is also regular. (Hint: Construct a regular expression for L^R from a regular expression for L .)
- (b) Prove that your regular expression from (a) represents L^R . (Hint: Use structural induction.)
- (c) Apply your construction to the α from Q1 to obtain a regular expression for $\mathcal{L}(\alpha)^R$.

Q5. Let $\Sigma = \{a, b\}$.

- (a) Prove that the set of all regular expressions over Σ is countable.
- (b) Prove that the set of all languages over Σ is uncountable.
- (c) Hence, prove that most languages over Σ cannot be represented by any regular expression.

Definition: deterministic finite automaton

A *deterministic finite automaton* (DFA) is a quintuple $M = (K, \Sigma, \delta, s, F)$ where:

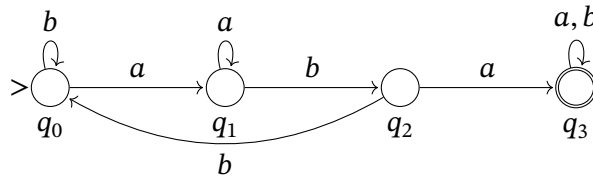
- K is a finite set of states,
- Σ is an alphabet,
- $s \in K$ is the initial state,
- $F \subseteq K$ is the set of final states, and
- $\delta: K \times \Sigma \rightarrow K$ is called the transition function.

Definition: configuration and computation

Let $M = (K, \Sigma, \delta, s, F)$ be a DFA. A *configuration* of M is an element of $K \times \Sigma^*$. We say that a configuration (q, w) *yields* (q', w') in one step, written $(q, w) \vdash_M (q', w')$, if $w = aw'$ for some $a \in \Sigma$ and $\delta(q, a) = q'$.

Let $\vdash_M^*$ denote the reflexive, transitive closure of $\vdash_M$. A string w is *accepted* by M if $(s, w) \vdash_M^* (q, e)$ for some $q \in F$. The language accepted by M , $L(M)$, is the set of all strings accepted by M .

Q6. Let M be the deterministic finite automaton below:



- Write down the full specification of M from the definition.
- Write out the full sequence of configurations attained during the operation $(q_0, baabab) \vdash_M^* (q_3, e)$, and confirm that $baabab \in L(M)$.
- Describe $L(M)$ in one sentence, in plain English.
- State and prove a necessary and sufficient condition on an arbitrary DFA M for $e \in L(M)$ to hold.

Q7. Let $\Sigma = \{a, b\}$.

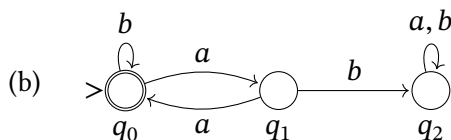
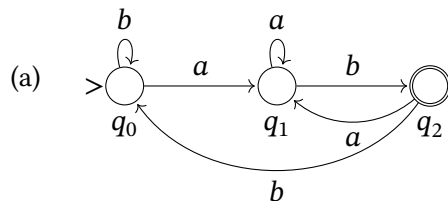
- Draw the state diagram for a DFA that accepts the language of all strings that do *not* contain a substring aab .
- What is the interpretation of each of the states in your DFA?

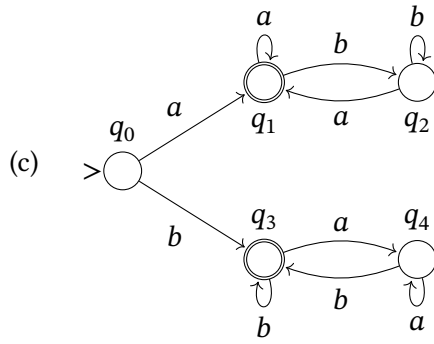
Q8. Let $M = (K, \Sigma, \delta, s, F)$ be a deterministic finite automaton where $K = \{q_0, q_1, q_2\}$, $\Sigma = \{a, b\}$, $s = q_0$, and $F = \{q_2\}$. The transition function is defined as follows:

$$\begin{array}{lll} \delta(q_0, a) = q_1 & \delta(q_1, a) = q_2 & \delta(q_2, a) = q_2 \\ \delta(q_0, b) = q_0 & \delta(q_1, b) = q_1 & \delta(q_2, b) = q_2 \end{array}$$

- Draw the state diagram for M .
- Describe $L(M)$ in one sentence, in plain English.

Q9. For each of the following deterministic finite automata over $\Sigma = \{a, b\}$, describe the language it accepts in a single, plain English sentence.





Q10. Let $\Sigma = \{a, b\}$. Let L be the language of all strings in which the number of a 's is even, and the number of b 's is a multiple of 3.

- Draw the state diagram of a DFA that accepts L .
- Explain how you could modify your answer to (a) to accept strings where the number of a 's is odd and the number of b 's has a remainder of 1 when divided by 3.

Q11. In Q5, we established that there are uncountably many languages but only countably many regular expressions.

- Argue informally that the set of all deterministic finite automata over an alphabet Σ is countably infinite. (Hint: How would you type the quintuple of any DFA into a plaintext file?)
- What does this mean for the set of all languages accepted by DFAs?

Q12. Let $M_1 = (K_1, \Sigma, \delta_1, s_1, F_1)$ and $M_2 = (K_2, \Sigma, \delta_2, s_2, F_2)$ be two DFAs. M_1 and M_2 are called *isomorphic* if there is a bijection $\varphi: K_1 \rightarrow K_2$ such that $\varphi(s_1) = s_2$, $\varphi(F_1) = F_2$, and $\varphi(\delta_1(q, a)) = \delta_2(\varphi(q), a)$ for all $q \in K_1$, $a \in \Sigma$.

- Write down two DFAs M_1 and M_2 over the alphabet $\{a, b\}$ with $K_1 \neq K_2$ that are isomorphic, and specify the bijection φ .
- Suppose that DFAs M_1 and M_2 accept the same language. Are they isomorphic? Give a short reason for your answer.

Q13. A *monoid* is a set M together with an operation $M \times M \rightarrow M$, $(a, b) \mapsto ab$ that satisfies two properties:

- Associativity:** $(ab)c = a(bc)$ for all $a, b, c \in M$.
- Identity:** There exists an $e \in M$ such that $ea = ae = a$ for all $a \in M$.

- Prove that Σ^* forms a monoid with string concatenation.
- Let Σ be a non-empty alphabet. Show that the set of all languages over Σ with the operation $(L_1, L_2) \mapsto \{uv : u \in L_1, v \in L_2\}$ forms a monoid. What is its identity element? What does $\emptyset$ do under this operation?